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$$\begin{aligned} & \lim_{x \rightarrow \infty} \left(\frac{2|x| + 1}{x} \right) \\ & \lim_{x \rightarrow +\infty} \left(\frac{2x + 1}{x} \right) \\ &= \lim_{x \rightarrow +\infty} \left(\frac{2x}{x} + \frac{1}{x} \right) \\ &= 2 + \frac{1}{+\infty} = 2 + 0 = 2 \\ & \lim_{x \rightarrow -\infty} \left(\frac{-2x + 1}{x} \right) \\ &= \lim_{x \rightarrow -\infty} \left(\frac{-2x}{x} + \frac{1}{x} \right) \\ &= -2 + \frac{1}{-\infty} = -2 - 0 = -2 \end{aligned}$$

References